Response to Reviewers

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An Explainable Stacking Framework for Multiclass Recorded Sleep-Status Classification

We thank the editor and reviewers. The revision changes both the analysis and the manuscript. We verified that "None" is a literal source-file category, identified extensive duplicate predictor profiles, replaced the single random hold-out analysis with repeated duplicate-aware validation, and released executable code and observation-level predictions. A bounded TreeSHAP diagnostic is generated only for the XGBoost base learner; it reports fold-to-fold stability and a local audit, not an explanation of the stack or clinical mechanism. The earlier 97.33% single-split result is not retained as a primary finding.

Reviewer 1

1. **Outcome coding.** The CSV is now read with missing-value conversion disabled and the code stops if an empty or undocumented outcome occurs. The audit finds 219 literal "None", 77 "Insomnia", and 78 "Sleep Apnea" labels, with no blank outcomes. We do not interpret "None" as a clinically adjudicated absence of disease (Methods, Data audit; Table 1).

2. **Primary configuration.** The revised primary analysis uses only RF+XGBoost stacking with a logistic meta-learner. SVM is removed. Soft voting uses exactly the same two base learners.

3. **Reproducibility.** `revised_analysis.py` is the executable analysis. It exports software versions, split/group assignments, fold metrics, reference labels, predictions, and probabilities. Preprocessing is refit inside every inner stacking fold.

4. **Validation and comparisons.** Two shuffled repetitions of five-fold StratifiedGroupKFold replace the single hold-out. Exact retained-predictor duplicates cannot cross a fold. Stacking, soft voting, both base learners, logistic regression, a shallow tree, and histogram gradient boosting use identical outer folds.

5. **Statistics.** The released predictions permit reproduction of all comparisons. Stacking minus soft-voting mean accuracy is -0.0031 (bootstrap 95% interval -0.0162 to 0.0070); the first-repeat out-of-fold McNemar result is $p = 0.6875$. No superiority claim remains.

6. **Explanations.** The prior SHAP/LIME/PCA/ALE artwork is not used as evidence for the stack. The executable revision generates a TreeSHAP diagnostic for the XGBoost base learner across five grouped outer-test folds, with a defined 10th–90th stability band. Its local audit releases the CSV row, true and predicted labels, all probabilities, duplicate group, and additivity residual. The figures explicitly do not explain the stack or establish clinical mechanisms.

Reviewer 2

1. We no longer claim novelty for stacking. The contribution is the audit and duplicate-aware reproducible evaluation protocol.

2. Risk-stratification, clinical-decision-support, and diagnostic claims were removed. The manuscript describes prediction of recorded source-file labels only.

3. The single 75-observation result is replaced by repeated grouped validation, variability, probability scores, calibration artefacts, and paired comparisons.
4. Original post-hoc claims are removed. The new XGBoost base-learner diagnostic is presented only as model-dependence and stability information, never as a biological mechanism.
5. Histogram gradient boosting was added as a contemporary tabular comparator.
6. The paper was rewritten, condensed, and regenerated from the revised results. The suggested traffic forecasting, image segmentation, and remote-sensing papers are not topical to sleep-health tabular modelling or this methodology, so they were not cited.

Reviewer 3

1. The Discussion explicitly separates internal benchmark discrimination from requirements for real clinical screening.
2. Results explain the small number of discordant predictions, the interval crossing zero, and limited paired-test power.
3. Rather than speculate about correlated-feature attribution rankings, the paper reports the fold-to-fold instability of the new base-learner TreeSHAP rankings.
4. The inconsistent RF+XGBoost+SVM robustness analysis is removed.
5. Limitations now state the potential classification bias and unknown clinical meaning of the literal "None" category.
6. The local diagnostic identifies its CSV row, group, true and predicted labels, probabilities, and additivity residual; the released CSV includes all feature contributions.
7. The paper recommends prospective cohorts, development-only tuning, calibration assessment, and external validation.
8. The suggested references were reviewed for relevance and were not added because they are outside the paper's topic.

Reviewer 4

1. We verified source outcomes and report 265 duplicate retained-predictor rows, 109 unique profiles, and 22 profiles with conflicting recorded outcomes. Group validation is now mandatory.
2. The supplied code generates every reported result and releases package versions, groups, splits, probabilities, metrics, and paired statistics.
3. Person ID is removed before group construction. Specifications are fixed; no test-fold tuning is performed.
4. Repeated grouped validation, simple baselines, Brier scores, and out-of-fold reliability curves were added.
5. Stale SHAP/LIME/ALE artwork is not used. The paper contains only executable TreeSHAP figures for the XGBoost base learner, with explicit stability and scope limits.

Reviewer 5

1. The raw import, literal-label check, cleaning, and outcome counts are implemented in `load_data()` and documented in Table 1.
2. The 97.33% 75-observation score is no longer used as evidence; ten duplicate-aware outer folds and their variation are reported.
3. Code, versions, seeds, group splits, metrics, and observation-level predictions are included in the repository.
4. The RF+XGBoost two-learner configuration is used consistently throughout; SVM has been removed.

Files supplied

- `revised_submission.tex`, `revised_references.bib`, and compiled manuscript PDF;
- this response and the editable `response_to_reviewers.md`;
- `revised_analysis.py`, `revised_stacking_cls.ipynb`, and `requirements.txt`;
- `revision_results/` containing metadata, exact splits, fold metrics, predictions/probabilities, paired statistics, and calibration artefacts.